

## Solution Requirements

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 27 June 2026                                      |
| <b>Team ID</b>       | XXXXXX                                            |
| <b>Project Name</b>  | EduGenie Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 4 Marks                                           |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category        | Requirement Description                                                                                                                       | Priority |
|------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | Authentication              | The application supports secure Google sign-in or email-based login to provide each learner with a personalized learning workspace.           | High     |
| 2    | Authorization Levels        | The system provides Student access for learning activities and an Admin role for managing content, users, and system settings.                | Medium   |
| 3    | External Interfaces         | The system integrates with the Google Gemini API to generate explanations, study plans, practice questions, summaries, and learning feedback. | High     |
| 4    | Learning Request Processing | The system accepts a learner's topic, question, level, and preferred learning style, then generates a relevant AI-assisted response.          | High     |
| 5    | Assessment & Reporting      | The system creates quizzes, evaluates submitted answers, and presents learning progress and performance feedback to the learner.              | High     |
| 6    | Business Rules              | A learner must provide a valid learning query or topic. AI responses must remain education-focused and clearly identify AI-generated content. | High     |

| S.No | Requirement Category              | Requirement Description                                                                                                                              | Priority |
|------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7    | Compliance to Laws or Regulations | The application protects student information, follows data-privacy principles, and uses the Gemini API in accordance with applicable platform terms. | Medium   |
| 8    | Other                             | The system stores learning history, saved topics, quiz scores, and feedback so learners can revisit material and track improvement.                  | Medium   |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category                  | Requirement Description                                                                                                | Target Metric / Acceptance Criteria                                   |
|------|-------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1    | Performance & Speed           | The system should generate learning assistance quickly and keep the interface responsive while requests are processed. | Average AI response time under 5 seconds                              |
| 2    | Scalability                   | The system should support a growing number of learners and be ready for future cloud deployment.                       | Supports at least 100 concurrent users in a future deployment         |
| 3    | Security & Data Privacy       | User credentials, learning history, and API communication should be securely handled.                                  | HTTPS communication; no API key exposed in the user interface         |
| 4    | Reliability & Availability    | The application should provide stable learning support with graceful handling of Gemini API failures.                  | System availability above 95% during testing                          |
| 5    | Usability & Accessibility     | The interface should be simple, student-friendly, and easy to navigate across common devices.                          | A learner can request help or begin a quiz within 3 interaction steps |
| 6    | Maintainability & Portability | The code should be modular, reusable, and straightforward to deploy in different environments.                         | Modular services with documented setup and API configuration          |